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Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 5	
TITLE : Collection Classes and StringBuffer	

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

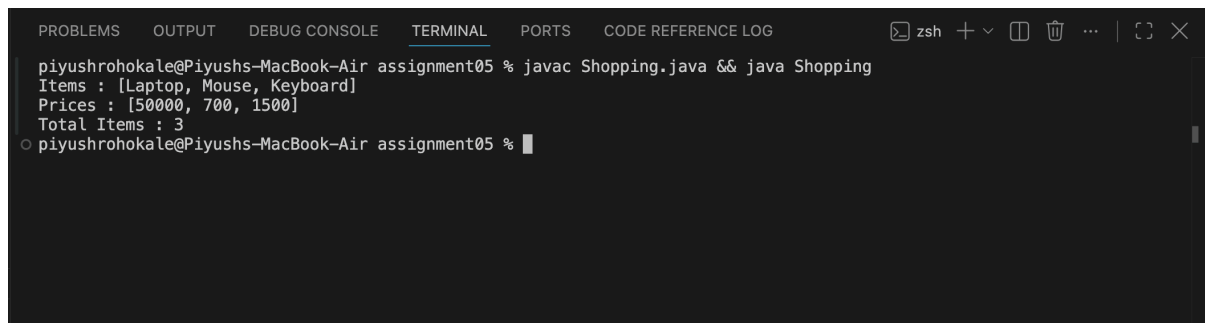
Programme

```

1  import java.util.ArrayList;
2  import java.util.Vector;
3
4  class Shopping
5  {
6
7      public static void main(String[] args)
8      {
9
10         ArrayList<String> items = new ArrayList<>();
11
12         items.add("Laptop");
13
14         items.add("Mouse");
15         items.add("Keyboard");
16
17         System.out.println("Items : " + items);
18
19         Vector<Integer> price = new Vector<>();
20
21         price.add(50000);
22
23         price.add(700);
24         price.add(1500);
25
26         System.out.println("Prices : " + price);
27
28         StringBuffer bill = new StringBuffer("Total Items : ");
29
30         bill.append(items.size());
31
32         System.out.println(bill);
33     }
34 }
35
36
37

```

Output

A screenshot of a terminal window with a dark background. The terminal shows the execution of a Java program. The prompt is 'piyushrohokale@Piyushs-MacBook-Air assignment05 %'. The command entered is 'javac Shopping.java && java Shopping'. The output of the program is displayed on the next line: 'Items : [Laptop, Mouse, Keyboard]', 'Prices : [50000, 700, 1500]', and 'Total Items : 3'. The prompt is then shown again with a cursor: 'piyushrohokale@Piyushs-MacBook-Air assignment05 %'. The terminal window has tabs at the top: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active), 'PORTS', and 'CODE REFERENCE LOG'. There are also icons for window management on the right side of the terminal header.

```
piyushrohokale@Piyushs-MacBook-Air assignment05 % javac Shopping.java && java Shopping
Items : [Laptop, Mouse, Keyboard]
Prices : [50000, 700, 1500]
Total Items : 3
piyushrohokale@Piyushs-MacBook-Air assignment05 %
```

Conclusion

The program successfully demonstrates the use of ArrayList , Vector , and StringBuffer in a shopping cart application. arraylist is used to store the product names dynamically , Vector is used to store the corresponding product prices , and stringbuffer is used to create the bill efficiently. This implementation shows how Java collection classes simplify dynamic data management and how stringbuffer enables efficient string manipulation.